COMPUTER SCIENCE TRIPOS Part IB 75%, Part II 50% – 2021 – Paper 7

5 Formal Models of Language (pjb48)

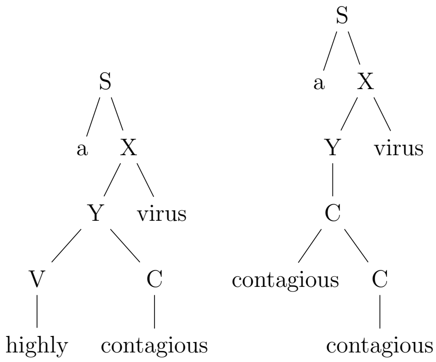
A linguist produces the grammar $G = (\mathcal{N}, \Sigma, S, \mathcal{P})$ where:

$$\begin{array}{lcl}
\mathcal{N} & = & \{\text{S, X, Y, V, C}\} \\
\Sigma & = & \{a, contagious, highly, virus\} \\
S & = & \text{S} \\
\mathcal{P} & = & \{\text{S} \rightarrow \text{a X, X} \rightarrow \text{Y } virus \mid virus\text{, Y} \rightarrow \text{V C} \mid \text{C,} \\
 & & \text{V} \rightarrow highly \text{ V} \mid highly\text{, C} \rightarrow contagious \text{ C} \mid contagious\}
\end{array}$$

context free grammars

(a) Draw all the trees with 4 leaves that can be derived from this grammar. [2 marks]

Answer:



[1 mark each]

PCFG

(b) Based on corpus data the linguist assigns probabilities to each rule in his grammar. Describe how the probability of a string is calculated from the rule probabilities. [2 marks]

Answer: The probability of a particular tree is the product of the probabilities of the rules that generated the tree [1 mark]. The probability of a string is the sum of all the parse trees that yield that string [1 mark].

A mathematician prefers to generate the strings of a language inductively. She defines a homomorphism: $\{(a, a), (c, contagious), (h, highly), (v, virus)\}$. She defines $L \subset \Sigma^*$ where $\Sigma = \{a, c, h, v\}$ using the following axioms and rules:

$$\frac{}{av}\ (\text{a1})$$

$$\frac{u_1 v}{u_1 c v}\ (\text{r1}) \text{ where } u_1 \in \Sigma^*$$

$$\frac{a c u_1}{a h c u_1}\ (\text{r2}) \text{ where } u_1 \in \Sigma^*$$

$$\frac{u_1 h u_2}{u_1 h h u_2}\ (\text{r3}) \text{ where } u_1, u_2 \in \Sigma^*$$

inductively defined languages

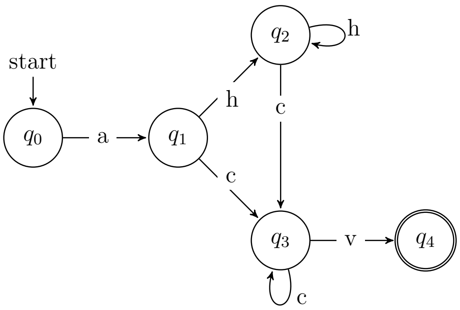
(*c*) Let $L_i = \{u \in L \mid \mathit{length}(u) \leq i\}$. Find all members of L_4 [3 marks]

Answer: $L_4 = \{av, acv$ [1 mark]$, accv$ [1 mark]$, ahcv$ [1 mark]$\}$

DFA

(*d*) Describe L as a regular expression and specify a Deterministic Finite Automaton, M, such that $L(M) = L$. [4 marks]

Answer: L contains strings matched by the regular expression $ah^*c^+v|av$ [1 mark] $M_1 = (\mathcal{Q}, \Sigma, \Delta, s, \mathcal{F})$.



where:

- $\mathcal{Q} = \{q_0, q_1, q_2, q_3\}$
- $\Sigma = \{a, c, h, v\}$
- $\Delta \subseteq \mathcal{Q} \times \Sigma \times \mathcal{Q}$ as shown in a transition table or diagram (partial or total function accepted)
- $s = q_0$
- $\mathcal{F} = \{q_3\}$

[1 mark for deterministic, 1 mark for c without h, 1 mark for recurrence]

entropy

(*e*) Provide an expression for the conditional entropy of X for L_5, where X is a random variable over Σ. A numerical value is not required. [5 marks]

Answer: $L_5 = \{av, acv, accv, ahcv, ahccv, ahhcv, acccv\}$ [1 mark]

For a 2nd order model we consider the immediate context. Let X_1 and X_2 refer to a sequence of events of X:

$$H(X_2|X_1) = \sum_{x_1\in\Sigma} p(x_1)H(X_2|X_1 = x_1)$$
$$= -\sum_{x_1\in\Sigma}\sum_{x_2\in\Sigma} p(x_1, x_2)\log_2 p(x_2|x_1)$$
$$= -\sum_{x_1\in\Sigma}\sum_{x_2\in\Sigma} p(x_2|x_1)p(x_1)\log_2 p(x_2|x_1) \text{ [1 mark]}$$

$p(a) = 7/28, p(c) = 10/28, p(v) = 7/28, p(h) = 4/28$ [1 mark]

$p(a|a) = 0, p(a|c) = 0, p(a|v) = 0, p(a|h) = 0$
$p(c|a) = 3/7, p(c|c) = 4/10, p(c|v) = 0, p(c|h) = 3/4$
$p(v|a) = 1/7, p(v|c) = 6/10, p(v|v) = 0, p(v|h) = 0$
$p(h|a) = 3/7, p(h|c) = 0, p(h|v) = 0, p(h|h) = 1/4$ [2 marks]

so $H(X) = -($
$p(c|a)p(a)\log_2 p(c|a) + p(c|c)p(c)\log_2 p(c|c) + p(c|h)p(h)\log_2 p(c|h)$
$+ p(v|a)p(a)\log_2 p(v|a) + p(v|c)p(c)\log_2 p(v|c)$
$+ p(h|a)p(a)\log_2 p(h|a) + p(h|h)p(h)\log_2 p(h|h))$ [1 mark]

This answer is fiddly so silly tiny mistakes are tolerated but workings should show that the student knows how to calculate the answer.

---

human language processing

(*f*) Suggest some hypotheses about human language processing that we could test based on the models mentioned in this question. Provide reasons for your hypotheses. [4 marks]

Answer: Based on the assumption that infrequent aspects of language will be more complicated to produce or comprehend one could hypothesise that a) reading time of a string is negatively correlated to the probability of that string as calculated by summing the probability of its PCFG trees; b) that your eye linger longer on characters sequences that are surprising as calculate by a second order model over the alphabet. Based on the assumption that linguistic constituency manifests cognitively one might hypothesis that the fully left or right linear trees produced by the mathematician's grammar do not allow grammatical substitution [1 mark for each reasonable hypothesis]
